



Merlina ARUL

Computer Science Engineering Student – Seeking Research Internship (2-3 months)

Computer science engineering student and current data engineer apprentice, seeking a 2-3 month research internship in an international environment. Interested in data engineering, software systems, and applied research.

Projects

MINI JIRA API - (Java/Spring Boot) :

REST API inspired by Jira for ticket management (creation, assignment, status, priorities). Layered architecture (Controller, Service, Repository) using Spring Boot. User, project, and task management via RESTful endpoints. Client-server communication using JSON. Demonstrates back-end system design and modular development.

SELF-SUSTAINING FARM (in Java) :

Simulation in Java of an autonomous farm where the user manages resources (water, food, fertilizer) of a field and entities (animals, plants, well) over multiple seasons. Object-oriented design with focus on entity interactions, resource constraints, and cyclic dependencies. Demonstrates modeling of complex systems and class architecture.

HUFFMAN CODING (in C) :

Implementation of the Huffman compression algorithm in C, including encoding, binary tree construction, and file decompression. Bit-level manipulation, binary file I/O, dynamic data structures, and memory management. Robustness handled (error cases, corrupted files). Demonstrates understanding of algorithmic optimization and data representation.

MASTERMIND (C with GTK) :

Development of a graphical interface for an existing Mastermind game using the GTK library. Designed a controller ensuring communication between the user interface and the game logic. Event-driven programming, user input handling, and display refresh management. Follows MVC architecture for clean separation of concerns.

SPACE INVADERS (in C) :

Complete implementation of Space Invaders in C using a pedagogical graphics library. Game mechanics, graphical rendering, collision detection, and projectile management. Reinforced core programming skills, game logic structuring, and code modularity. Focus on real-time event handling and state management.

Work experiences

Data Engineer apprentice

BNP Paribas, Paris Since September 2025

- Developed data purge processes with Spring Boot and Spring Batch
- Wrote SQL queries for extraction, validation, and maintenance
- Configured batch jobs with YAML and Jenkins
- Contributed to automated monitoring workflows
- Used Python and Airflow for orchestration
- Implemented email alerts for monitoring
- Working in an agile development environment

Crew member (part-time)

La Mie Câline, Meaux From January 2024 to September 2025

- Customer communication
- Team coordination in a fast-paced environment
- Adaptability and problem-solving

Education

Engineering Program – Computer Science Major

Sup Galilée, Paris Since September 2024

Integrated Preparatory Program in Engineering

Sup Galilée, Paris From September 2022 to June 2024

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Links :

in linkedin.com/in/merlinaarul
🐙 github.com/merlinaarul

Technical skills :

Programming languages :

- **Python** – Good proficiency
- **Java (Spring Boot/ Spring Batch)** – Solid knowledge
- **C** – Good proficiency, strong understanding of data structures
- **SQL** – Solid basics
- **MATLAB** – Good proficiency
- **HTML/CSS** – Basic knowledge

DevOps & Tools:

- Git
- Docker (containerization)
- Flyway (database migrations)
- Apache Airflow

Databases

- Relational modeling, conceptual design (ERD)

Networking

- Wireshark, ping, TCP/IP model

Operating Systems

- **Linux** – Comfortable with Linux environment (terminal, basic commands, file/process management)

Algorithms

- Complexity analysis, graph algorithms (DFS/BFS)

Languages

French
Native

English
Advanced

Tamil
Mother tongue

Soft skills

Team spirit

Self-reliance

Adaptability

Initiative

Interests :

Swimming, running, reading